

Multimedia Computing

By

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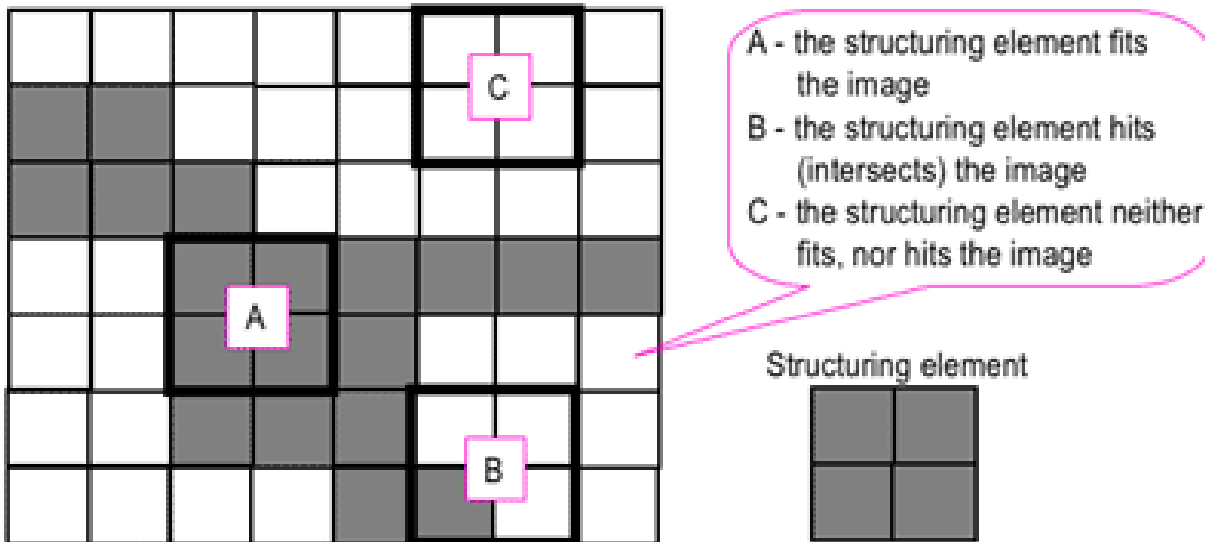
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Content

- ▶ Chapter - 9 (Gonzalez books)
- ▶ Morphological operation
- ▶ Dilation
- ▶ Erosion
- ▶ Opening
- ▶ Closing

Basic concepts

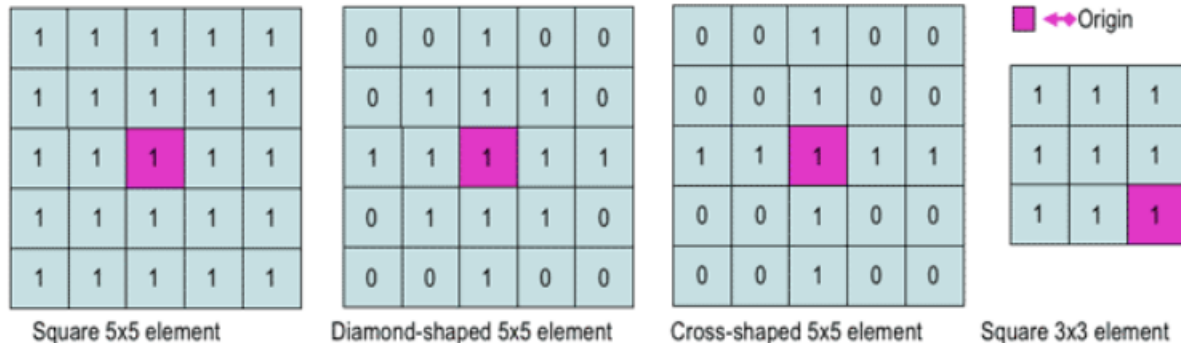
- **Morphological image processing** is a collection of non-linear operations related to the shape or morphology of features in an image.



Probing of an image with a structuring element
(white and grey pixels have zero and non-zero values, respectively).

Structuring element

- ▶ The **structuring element** is a small binary image, i.e. a small matrix of pixels, each with a value of zero or one:
- The matrix dimensions specify the *size* of the structuring element.
- The pattern of ones and zeros specifies the *shape* of the structuring element.
- An *origin* of the structuring element is usually one of its pixels, although generally the origin can be outside the structuring element.



Examples of simple structuring elements.

Morphological image processing

- ▶ A common practice is to have odd dimensions of the structuring matrix and the origin defined as the centre of the matrix. Structuring elements play in morphological image processing the same role as convolution kernels in linear image filtering.

Morphological image processing

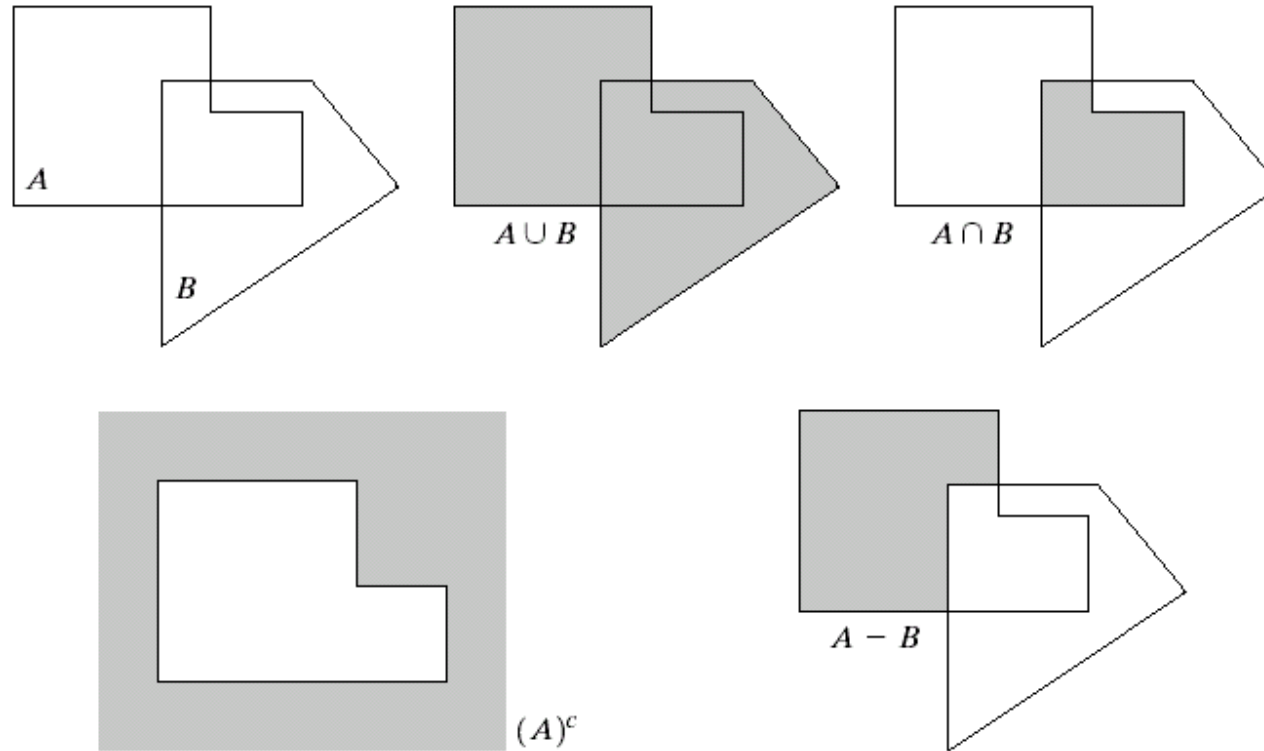
- ▶ When a structuring element is placed in a binary image, each of its pixels is associated with the corresponding pixel of the neighborhood under the structuring element. The structuring element is said to **fit** the image if, for each of its pixels set to 1, the corresponding image pixel is also 1. Similarly, a structuring element is said to **hit**, or intersect, an image if, at least for one of its pixels set to 1 the corresponding image pixel is also 1.



Fitting and hitting of a binary image with structuring elements s_1 and s_2 .

Zero-valued pixels of the structuring element are ignored, i.e. indicate points where the corresponding image value is irrelevant.

Some Basic Concepts from Set Theory

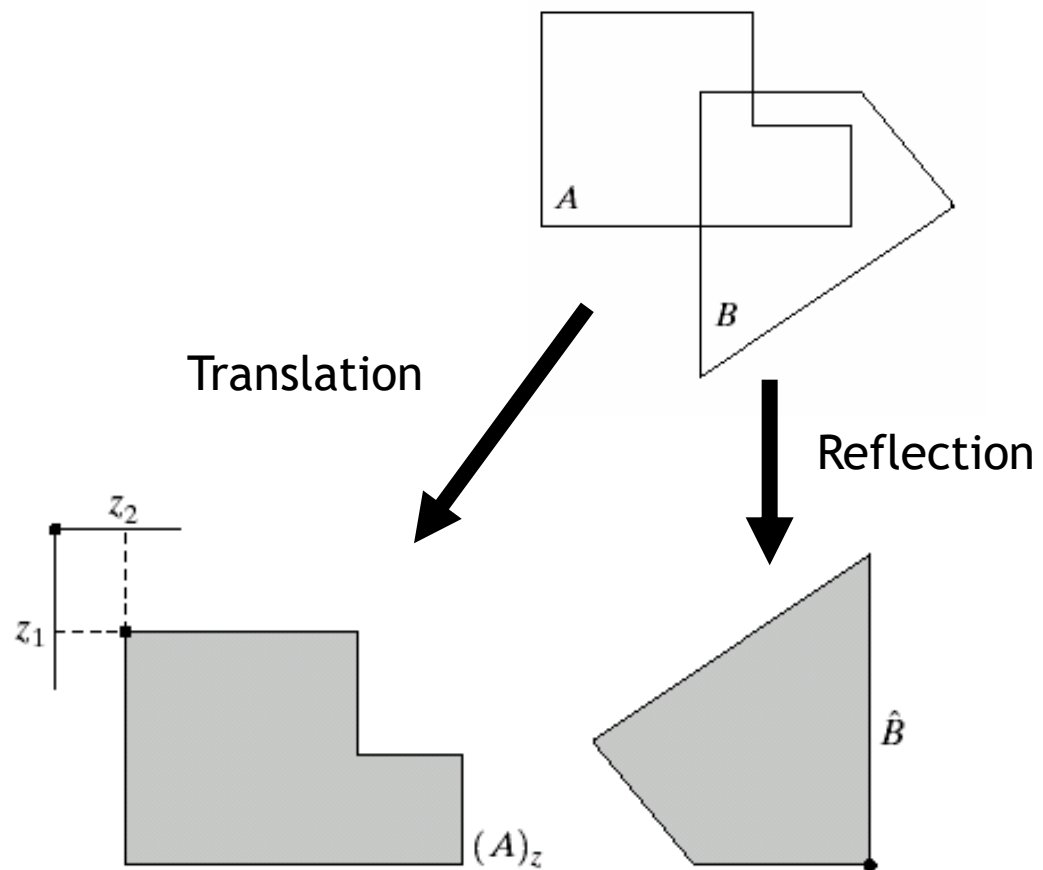


a	b	c
d	e	

FIGURE 9.1

(a) Two sets A and B . (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B .

Some Basic Concepts from Set Theory



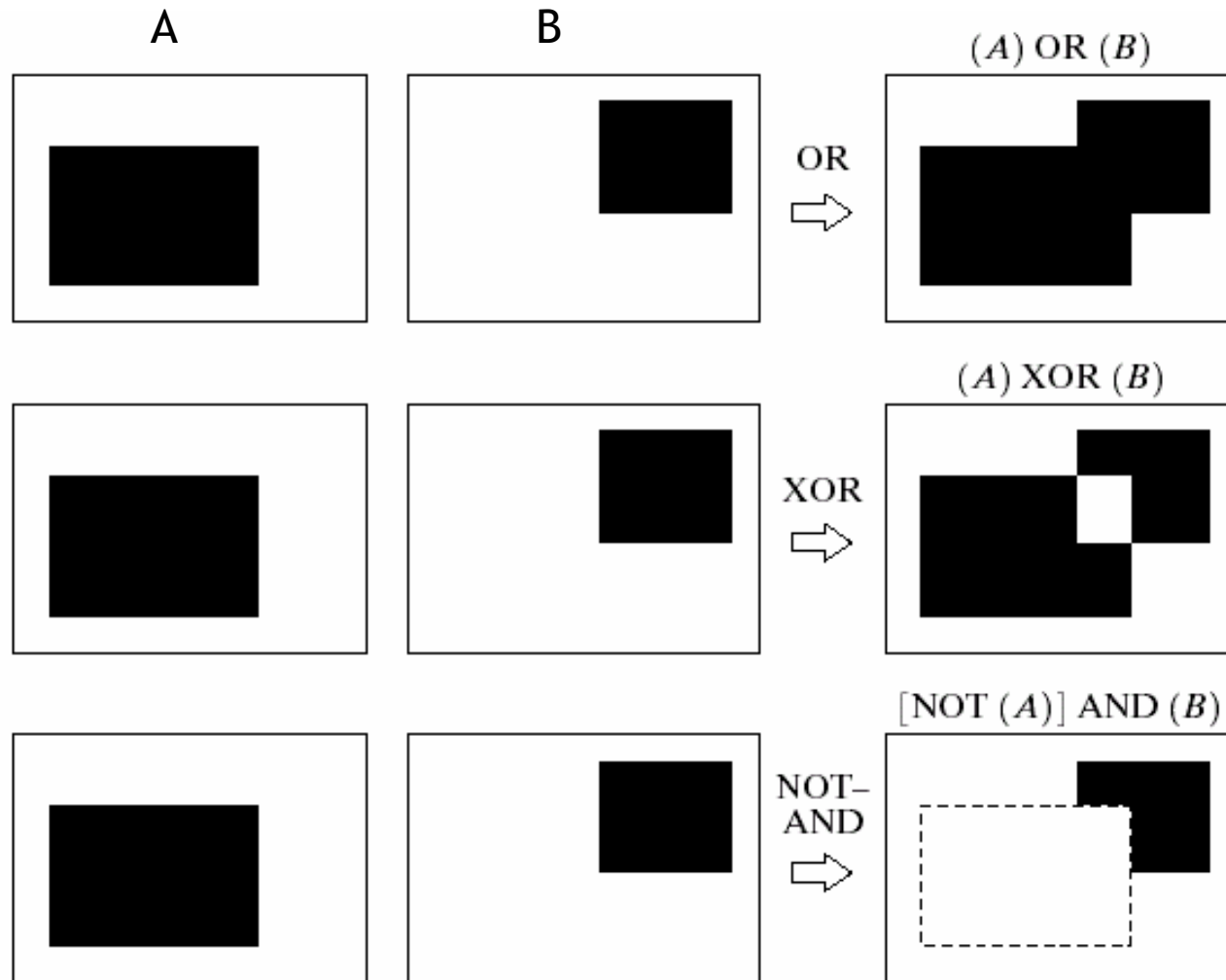
a b

FIGURE 9.2

(a) Translation of A by z .

(b) Reflection of B . The sets A and B are from Fig. 9.1.

Logic Operations Involving Binary Images



Basic concept

Morphological operations rely only on the relative ordering of pixel values, not on their numerical values, and therefore are especially suited to the processing of binary images. Morphological operations can also be applied to grayscale images such that their light transfer functions are unknown and therefore their absolute pixel values are of no or minor interest.

Basic Morphological operation

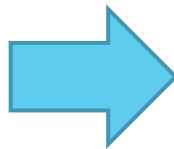
- ▶ A set of operations that process images based on shapes. Morphological operations apply a *structuring element* to an input image and generate an output image.
- ▶ The most basic morphological operations are two: **Erosion and Dilation**. They have a wide array of uses, i.e. :
 - ▶ Removing noise
 - ▶ Isolation of individual elements and joining disparate elements in an image.
 - ▶ Finding of intensity bumps or holes in an image

Dilation

This operations consists of convoluting an image A with some kernel (B), which can have any shape or size, usually a square or circle.

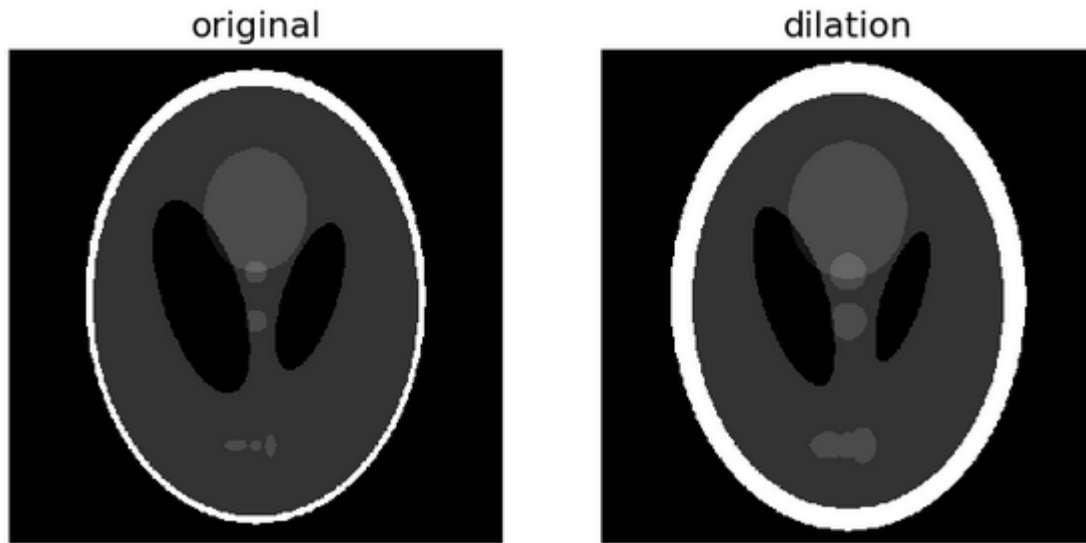
The kernel B has a defined *anchor point*, usually being the center of the kernel.

As the kernel B is scanned over the image, we compute the maximal pixel value overlapped by and replace the image pixel in the anchor point position with that maximal value. As you can deduce, this maximizing operation causes bright regions within an image to “grow therefore name is dilation



Dilation

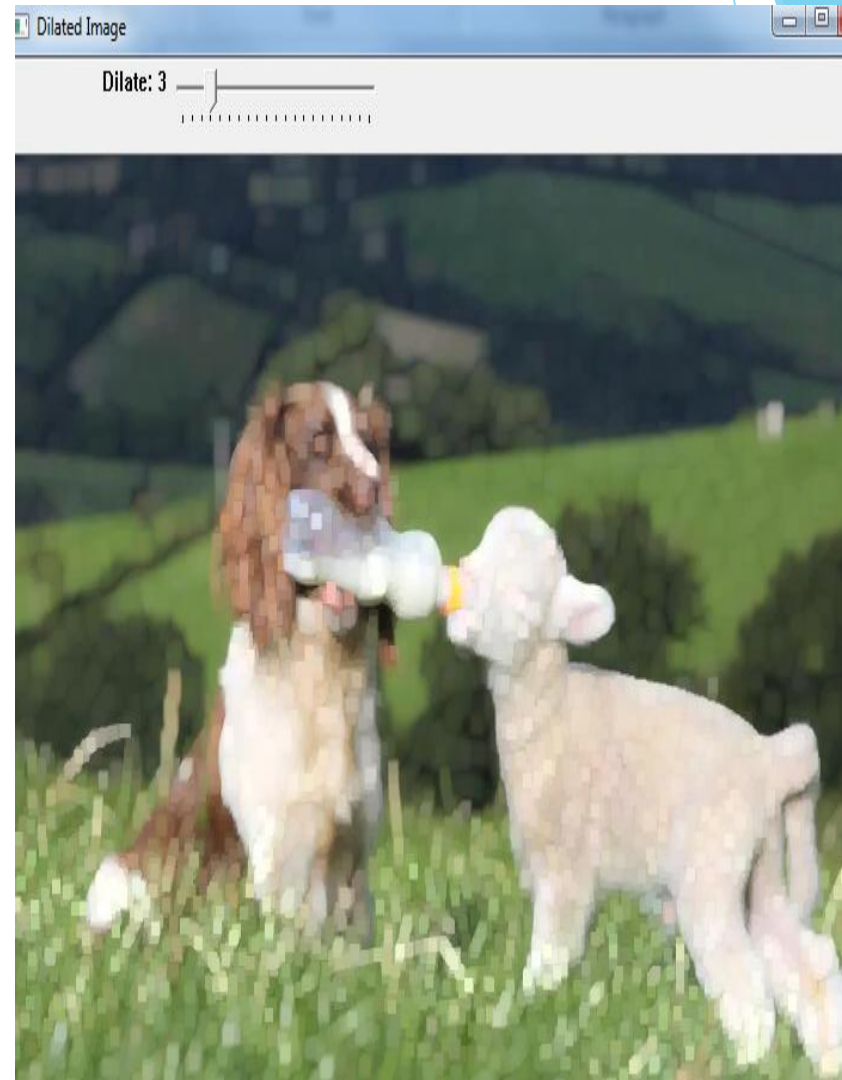
Morphological dilation sets a pixel at (i, j) to the *maximum over all pixels in the neighborhood centered at (i, j)* . Dilation enlarges bright regions and shrinks dark regions



Dilation



Dilation



Erosion



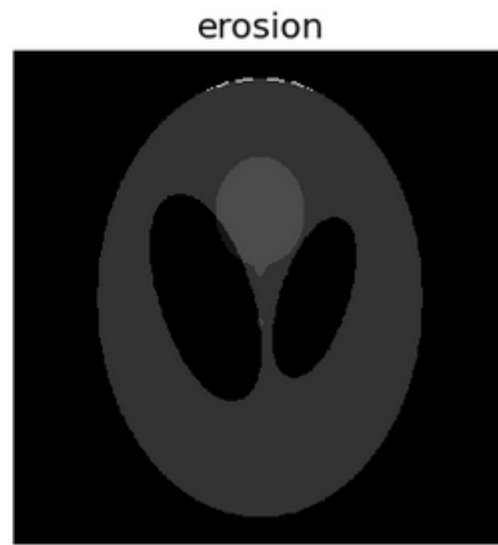
Common Names: Erode, Shrink, Reduce

The basic effect of the operator on a binary image is to erode away the boundaries of regions of foreground pixels (*i.e.* white pixels, typically). Thus areas of foreground pixels shrink in size, and holes within those areas become larger.

How it works

- ▶ The erosion operator takes two pieces of data as inputs.
- ▶ The second is a set of coordinate points known as a structuring element (also known as a *kernel*). It is this structuring element that determines the precise effect of the erosion on the input image.

Erosion

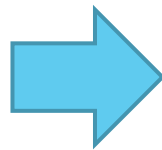


Erosion

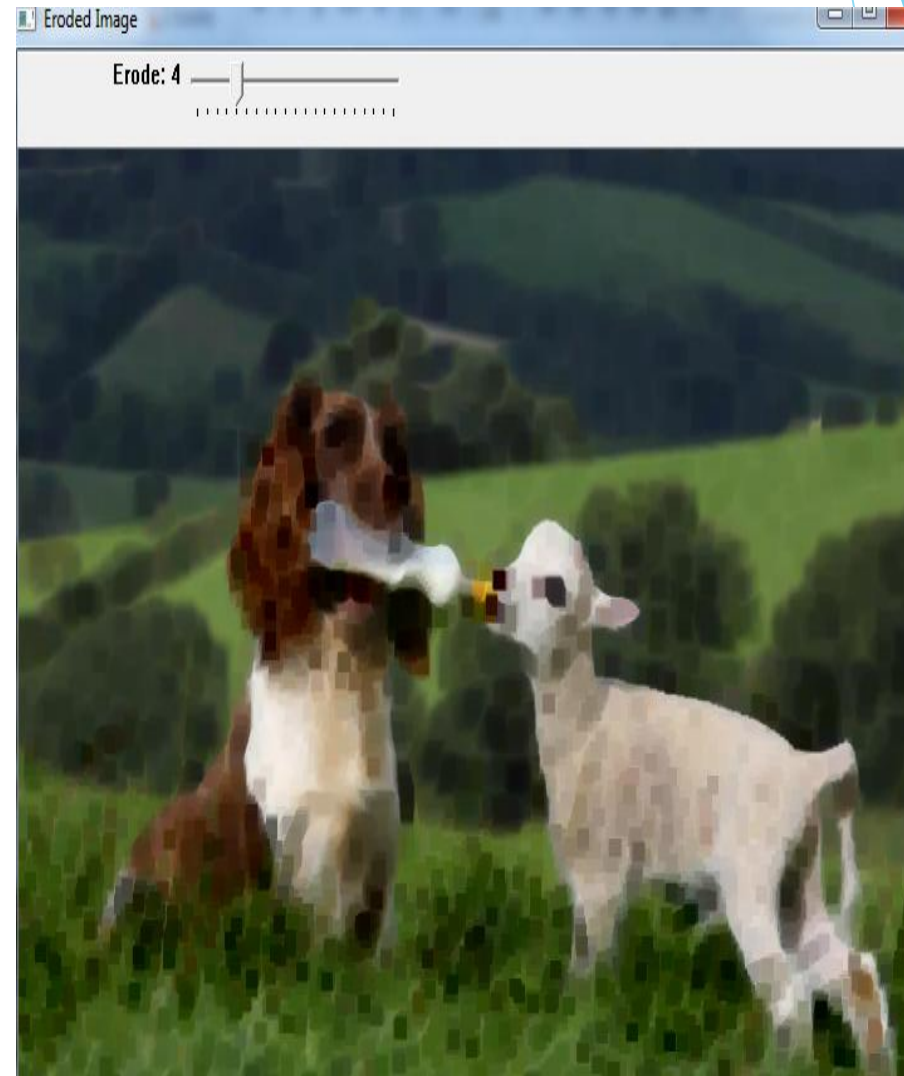
This operation is the sister of dilation. What this does is to compute a local minimum over the area of the kernel.

As the kernel B is scanned over the image, we compute the minimal pixel value overlapped by B and replace the image pixel under the anchor point with that minimal value.

erode the original image (shown below). You can see in the result below that the bright areas of the image (the background, apparently), get thinner, whereas the dark zones gets bigger.



Erosion



Dilation and Erosion

- ▶ With A and B as sets in Z^2 , the dilation of A by B is defined as

$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \phi\}$$

where $\hat{B}$: the reflection of B about its origin and shifting this reflection by z

- ▶ The dilation of A by B is the set of all displacements, z , such that $\hat{B}$ and A **overlap by at least one element**. Thus,

$$A \oplus B = \{z \mid [(\hat{B})_z \cap A] \subseteq A\}$$

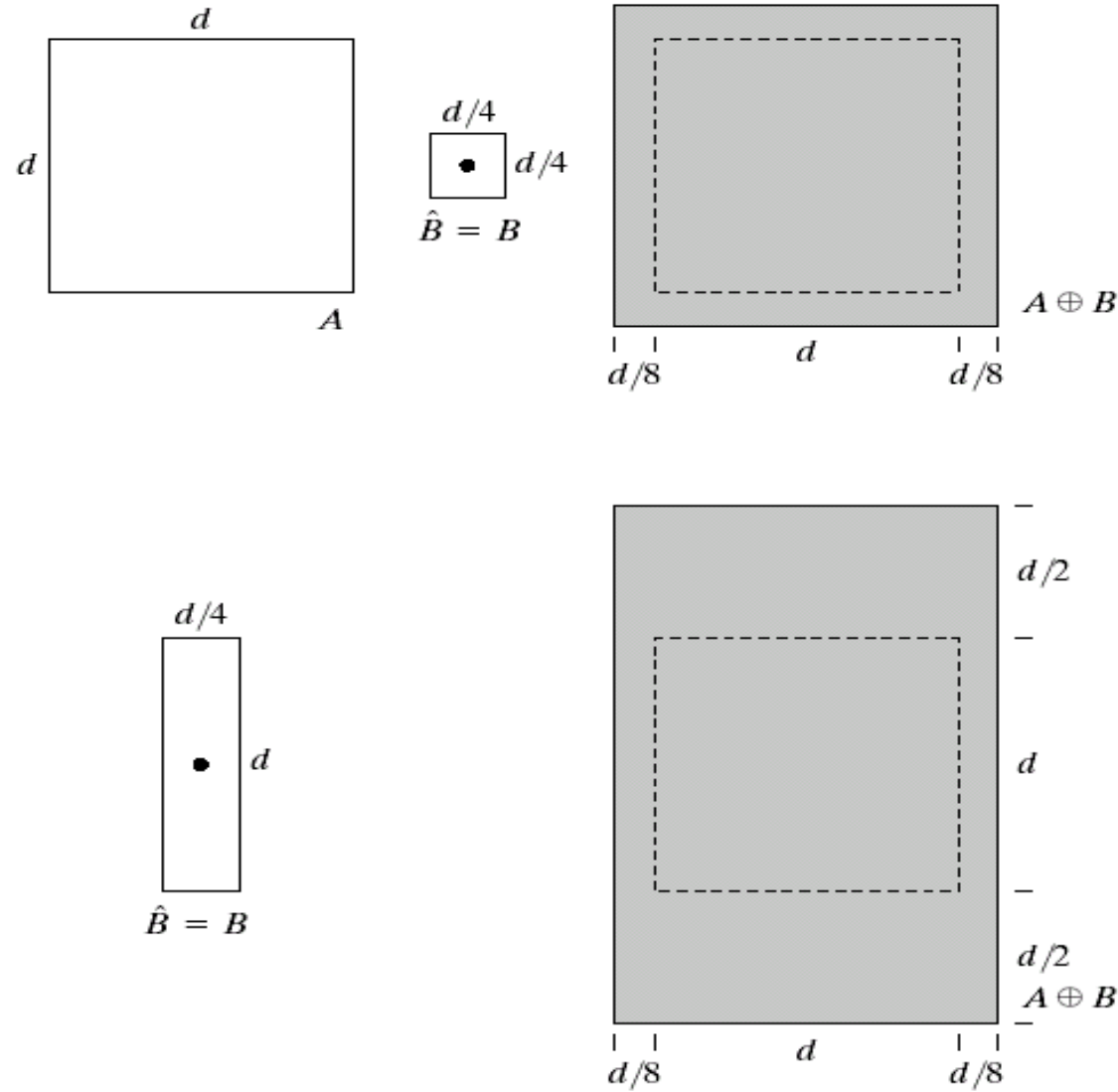
- ▶ Set B is referred to as the **structuring element** in dilation.

Dilation and Erosion Dilation

a	b	c
d		e

FIGURE 9.4

- (a) Set A .
 (b) Square structuring element (dot is the center).
 (c) Dilation of A by B , shown shaded.
 (d) Elongated structuring element.
 (e) Dilation of A using this element.



Dilation and Erosion

Dilation Example

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



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0	1	0
1	1	1
0	1	0

Structuring element

a c
b

FIGURE 9.5

(a) Sample text of poor resolution with broken characters (magnified view).
(b) Structuring element.
(c) Dilation of (a) by (b). Broken segments were joined.

Dilation and Erosion

Erosion

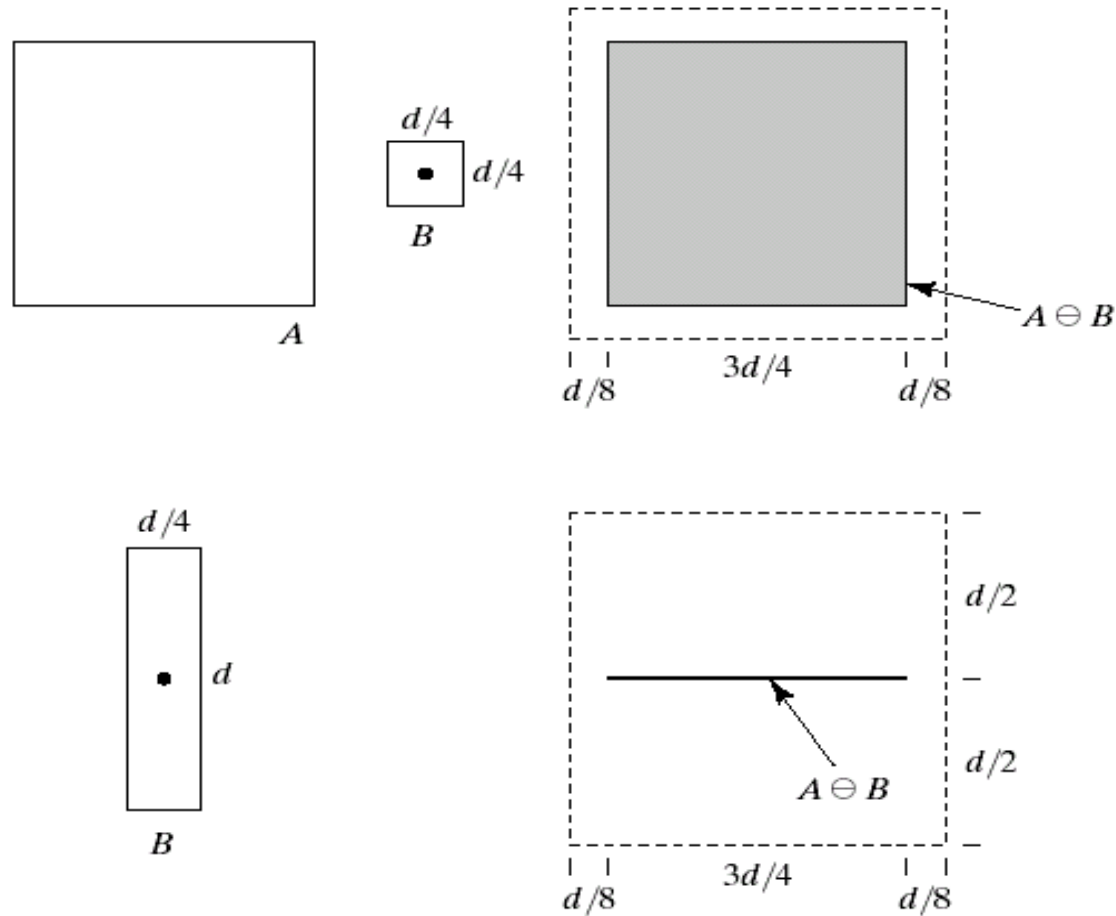
- ▶ For sets A and B in Z^2 , the erosion of A by B is defined as
$$A \ominus B = \{z \mid (\hat{B})_z \subseteq A\}$$
where $\hat{B}$: the reflection of B about its origin and shifting this reflection by z
- ▶ The erosion of A by B is the set of all points z , such that B , translated by z is contained in A .

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

$$\begin{aligned}(A \ominus B)^c &= \{z \mid (B)_z \subseteq A\}^c \\ &= \{z \mid (B)_z \subseteq A^c = \phi\}^c \\ &= \{z \mid (B)_z \subseteq A^c \neq \phi\} = A^c \oplus \hat{B}\end{aligned}$$

Dilation and Erosion

Erosion

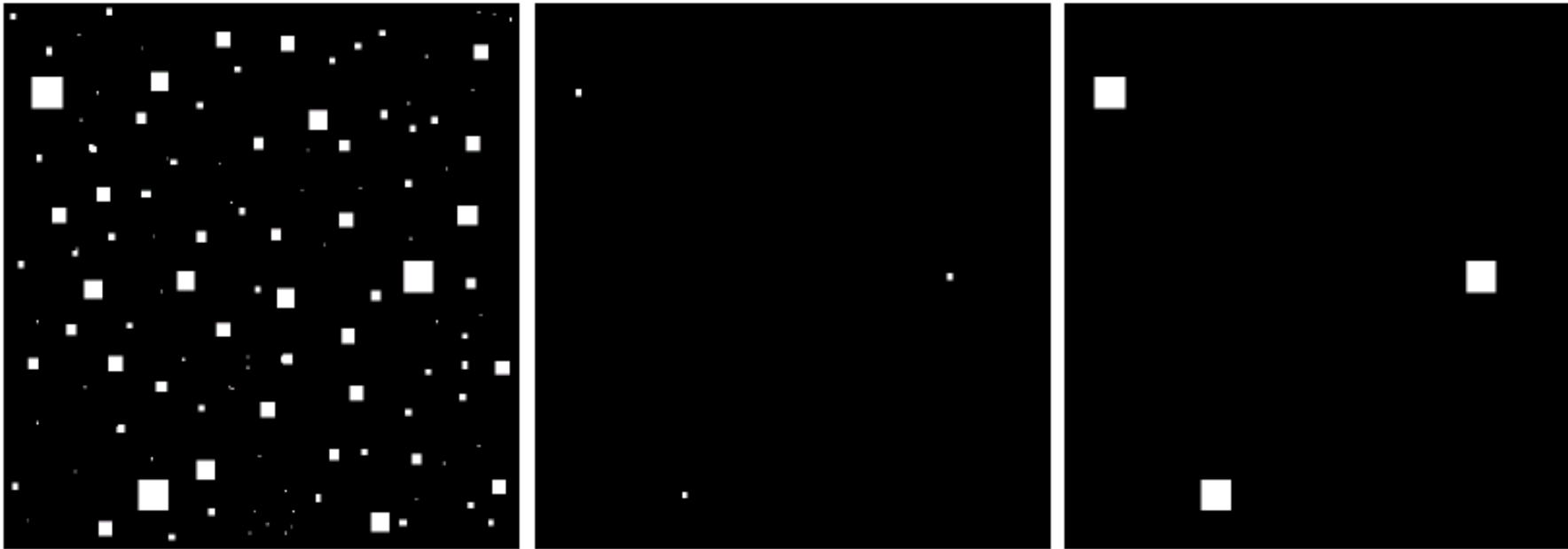


a	b	c
d		e

FIGURE 9.6 (a) Set A . (b) Square structuring element. (c) Erosion of A by B , shown shaded. (d) Elongated structuring element. (e) Erosion of A using this element.

Dilation and Erosion

Erosion Example



a b c

FIGURE 9.7 (a) Image of squares of size 1, 3, 5, 7, 9, and 15 pixels on the side. (b) Erosion of (a) with a square structuring element of 1's, 13 pixels on the side. (c) Dilation of (b) with the same structuring element.

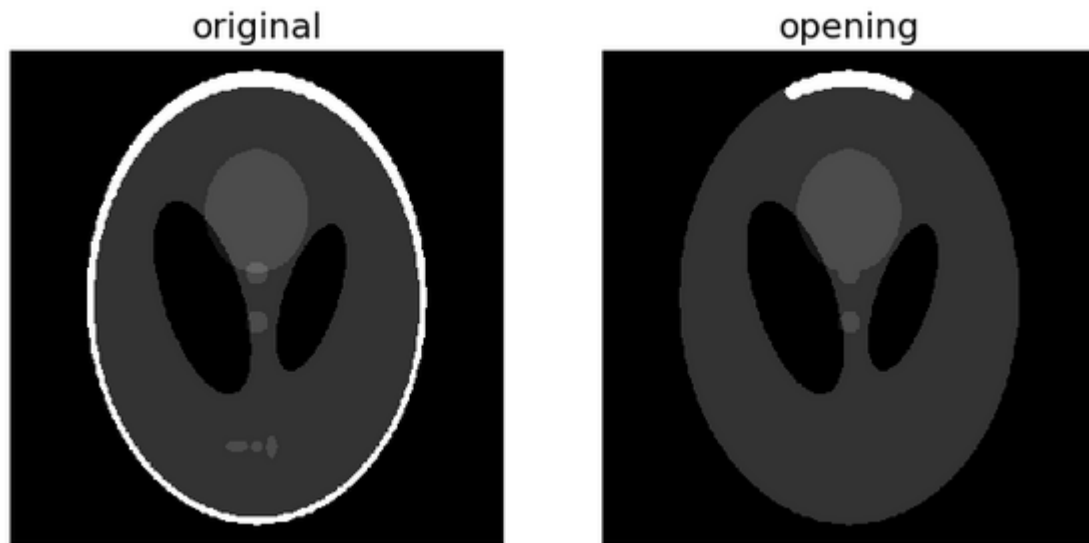
Opening

- ▶ It is obtained by the erosion of an image followed by a dilation.

The basic effect of an opening is somewhat like erosion in that it tends to remove some of the foreground (bright) pixels from the edges of regions of foreground pixels.

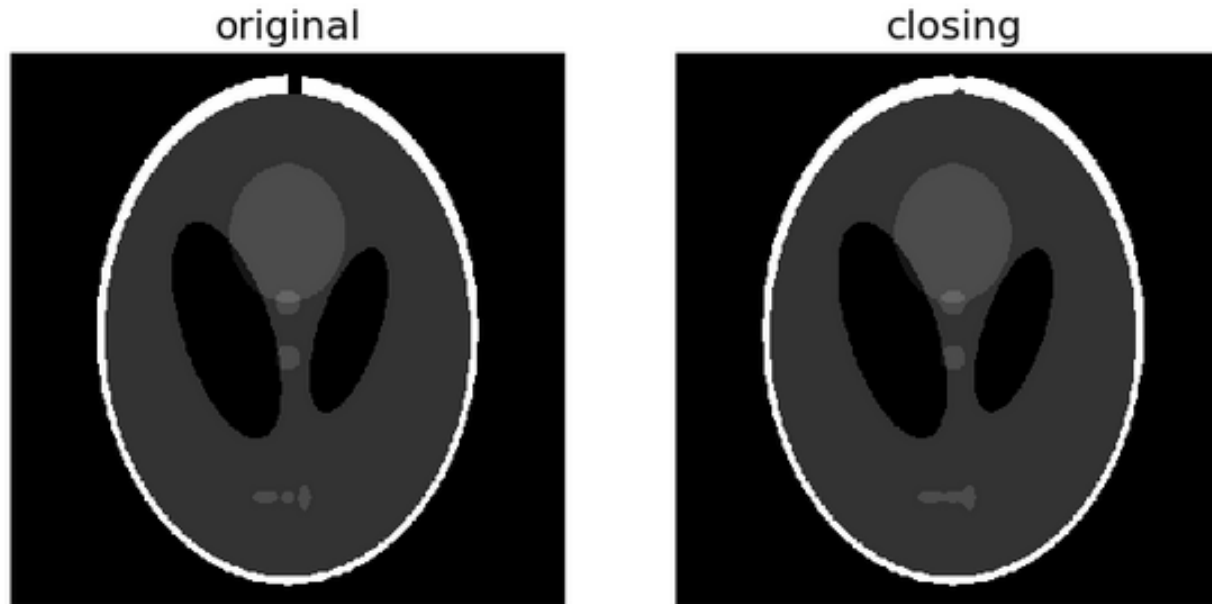
Opening can remove small bright spots and connect small dark cracks.

Opening



Closing

- ▶ Morphological closing on an image is defined as a *dilation* followed by an *erosion*. Closing can remove small dark spots and connect small bright cracks.



Opening and Closing

▶ Opening:

- ▶ smooth the contour of an object, break narrow isthmuses, and eliminate thin protrusions.
- ▶ The opening A by B is the erosion of A by B , followed by a dilation of the result by B

$$A \circ B = (A \ominus B) \oplus B$$

$$A \circ B = \cup \{(B)_z \mid (B)_z \subseteq A\}$$

▶ Closing:

- ▶ smooth sections of contours but it generally fuses narrow breaks and long thin gulfs, eliminates small holes, and fills gaps in the contour.

$$A \bullet B = (A \oplus B) \ominus B$$

Opening and Closing

Opening

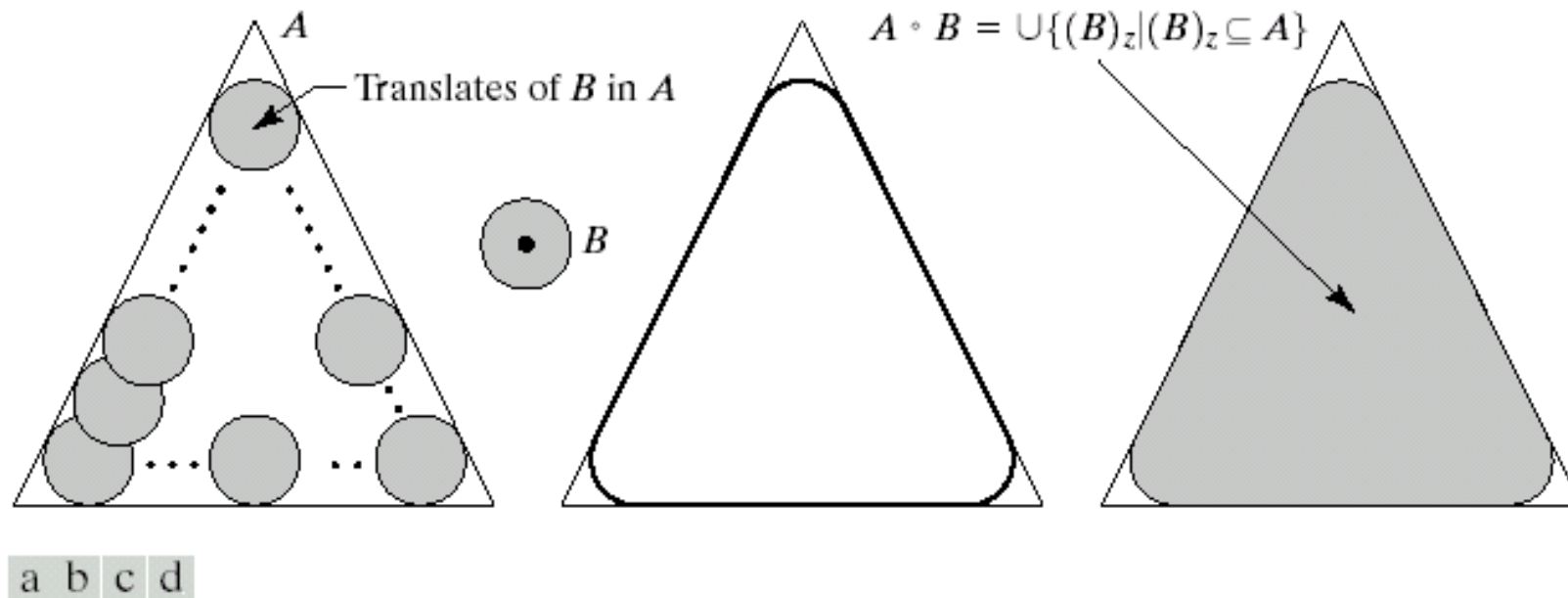


FIGURE 9.8 (a) Structuring element B "rolling" along the inner boundary of A (the dot indicates the origin of B). (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded).

Opening and Closing

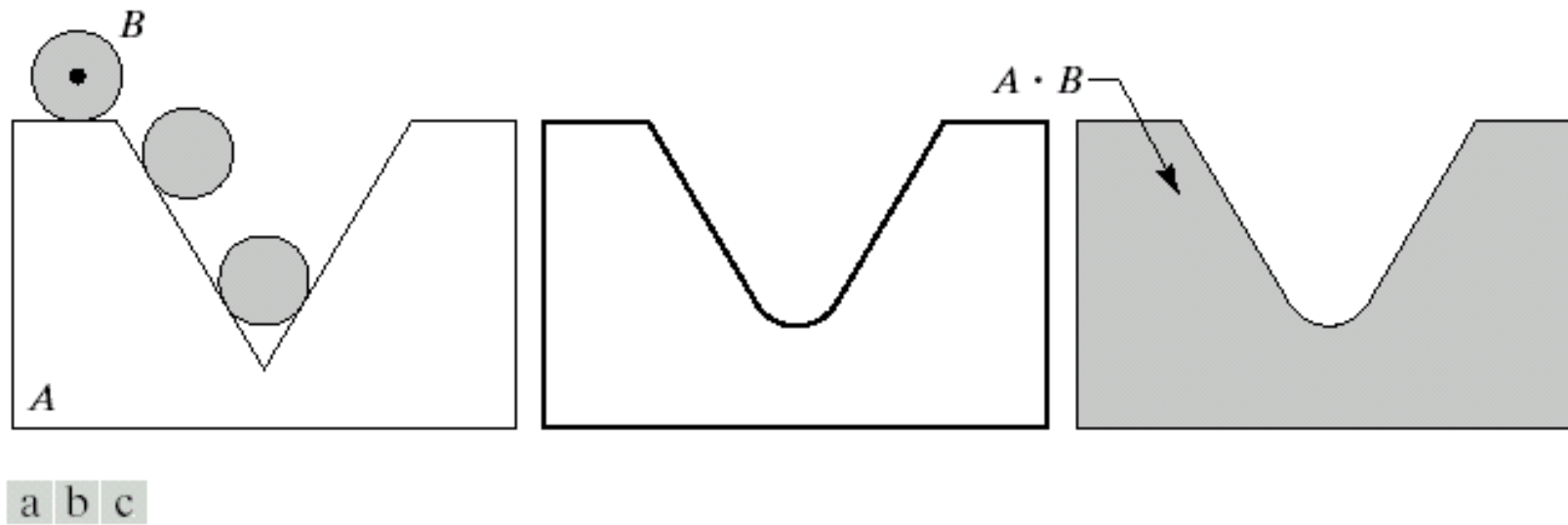


FIGURE 9.9 (a) Structuring element B “rolling” on the outer boundary of set A . (b) Heavy line is the outer boundary of the closing. (c) Complete closing (shaded).

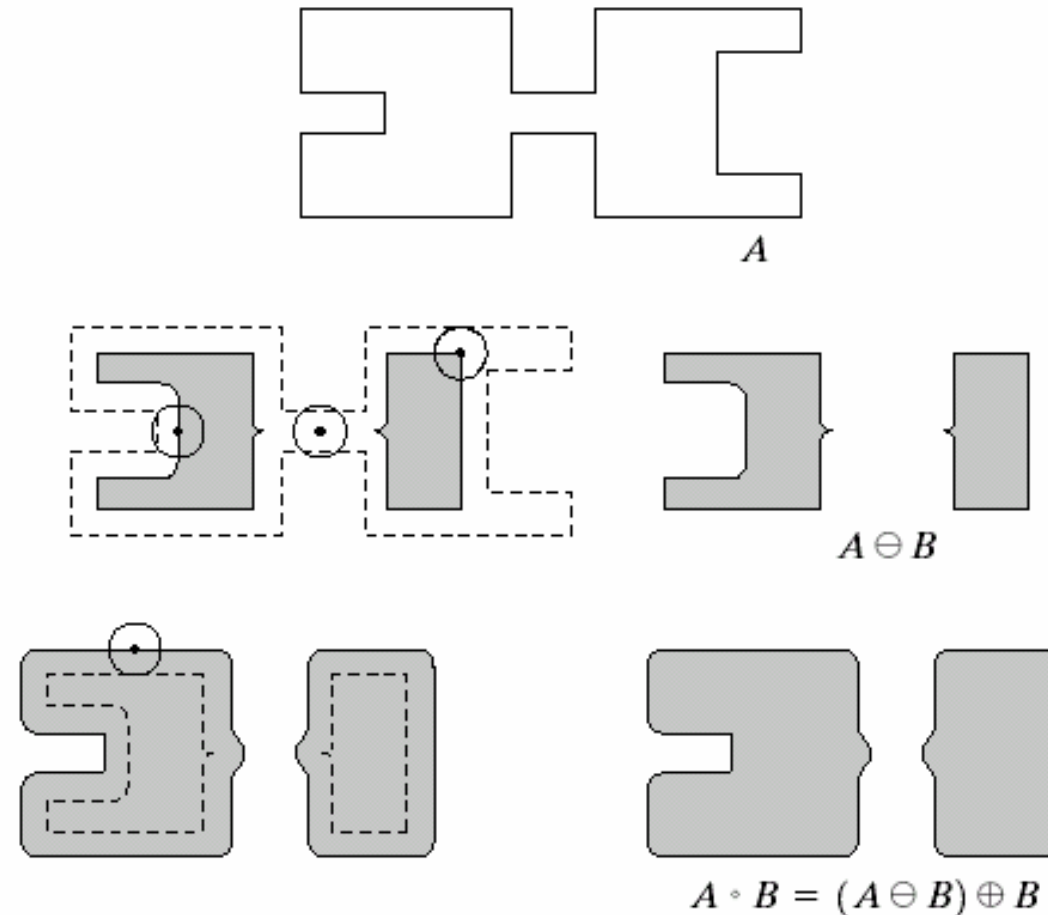
Opening and Closing

Opening Example

a
b c
d e
f g
h i

FIGURE 9.10

Morphological opening and closing. The structuring element is the small circle shown in various positions in (b). The dark dot is the center of the structuring element.

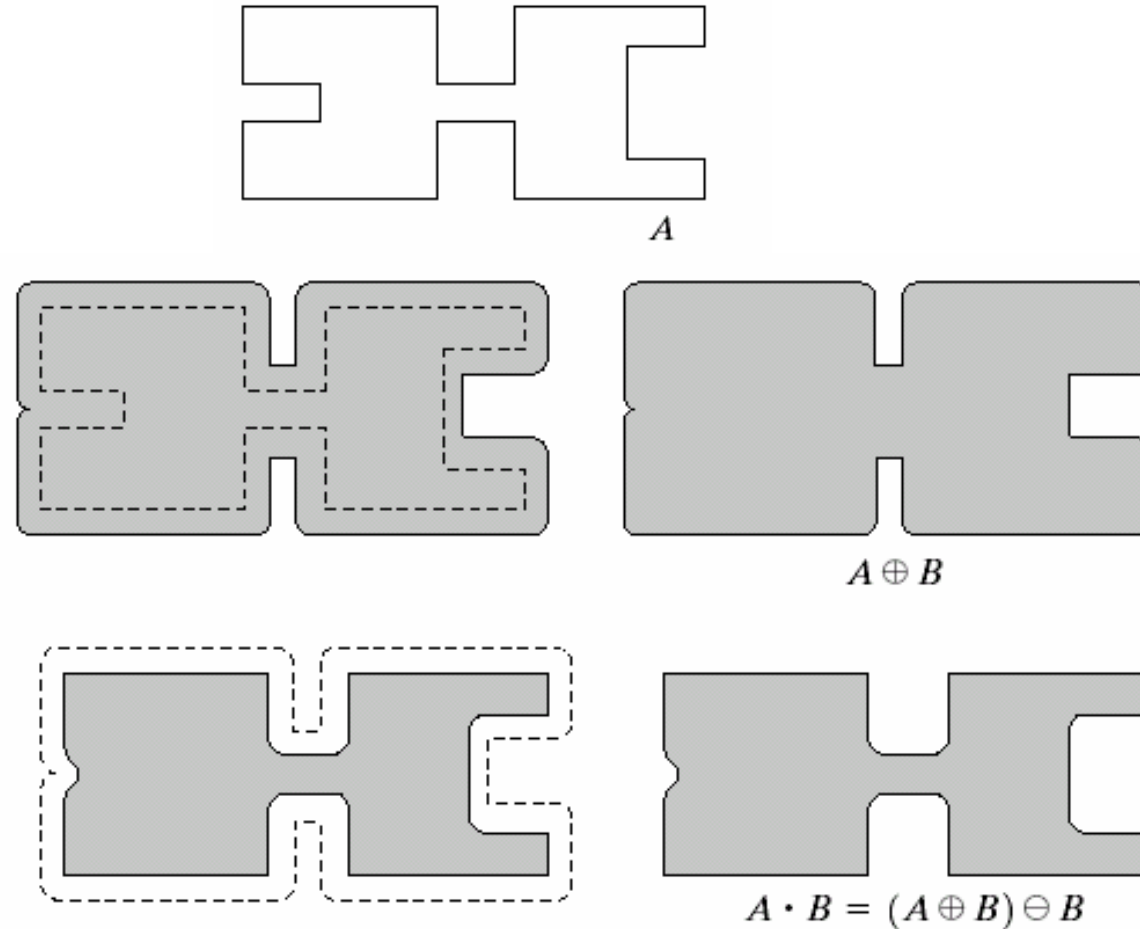


Opening and Closing

Closing Example

a
b c
d e
f g
h i

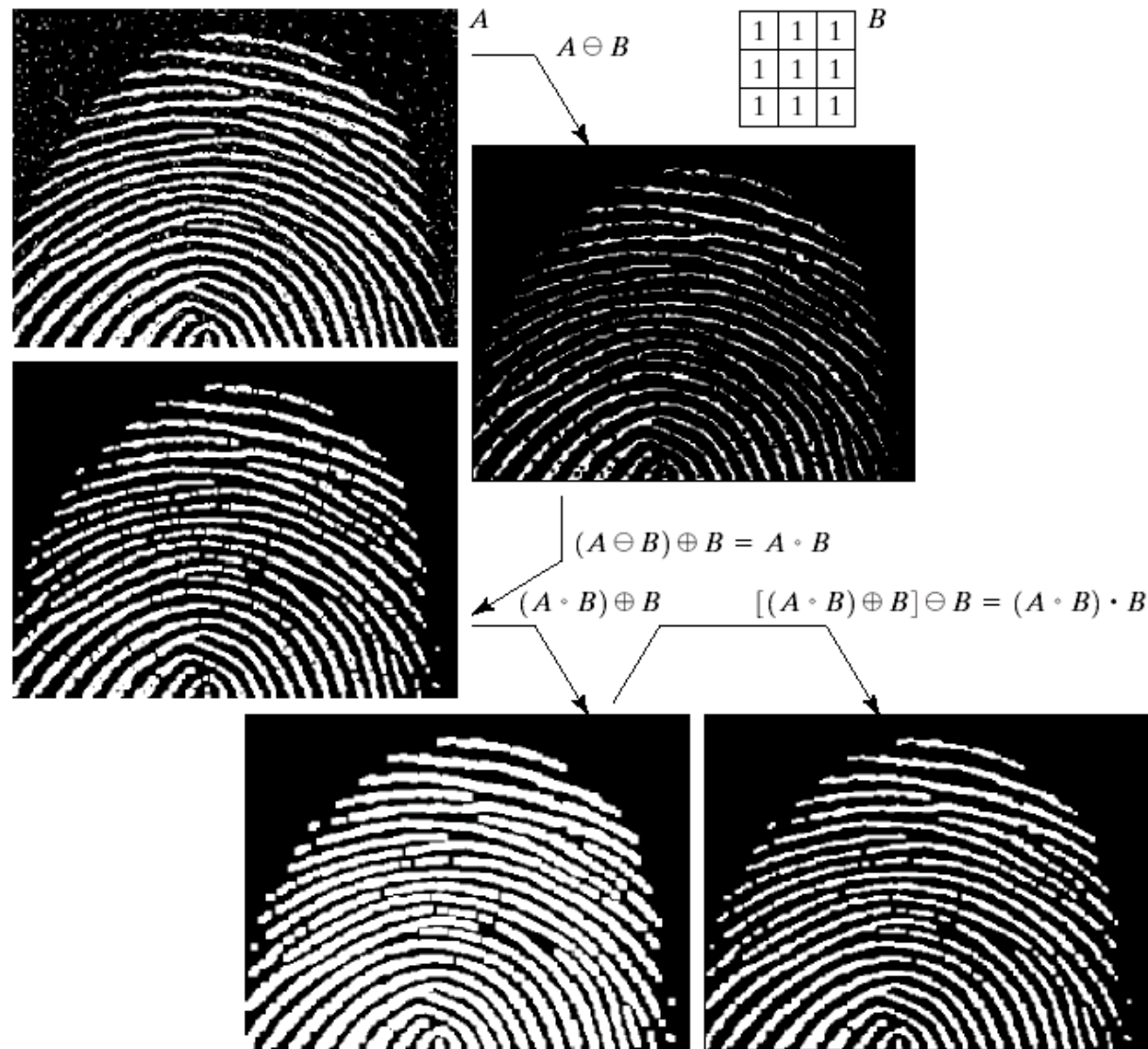
FIGURE 9.10
Morphological opening and closing. The structuring element is the small circle shown in various positions in (b). The dark dot is the center of the structuring element.



Opening and Closing

- ▶ The opening operation satisfies the following properties:
 - ▶ $A \circ B$ is a subset (subimage) of A .
 - ▶ If C is a subset of D , then $C \circ B$ is a subset of $D \circ B$.
 - ▶ $(A \circ B) \circ B = A \circ B$
- ▶ The closing operation satisfies the following properties:
 - ▶ A is a subset (subimage) of $A \bullet B$.
 - ▶ If C is a subset of D , then $C \bullet B$ is a subset of $D \bullet B$.
 - ▶ $(A \bullet B) \bullet B = A \bullet B$

Opening and Closing



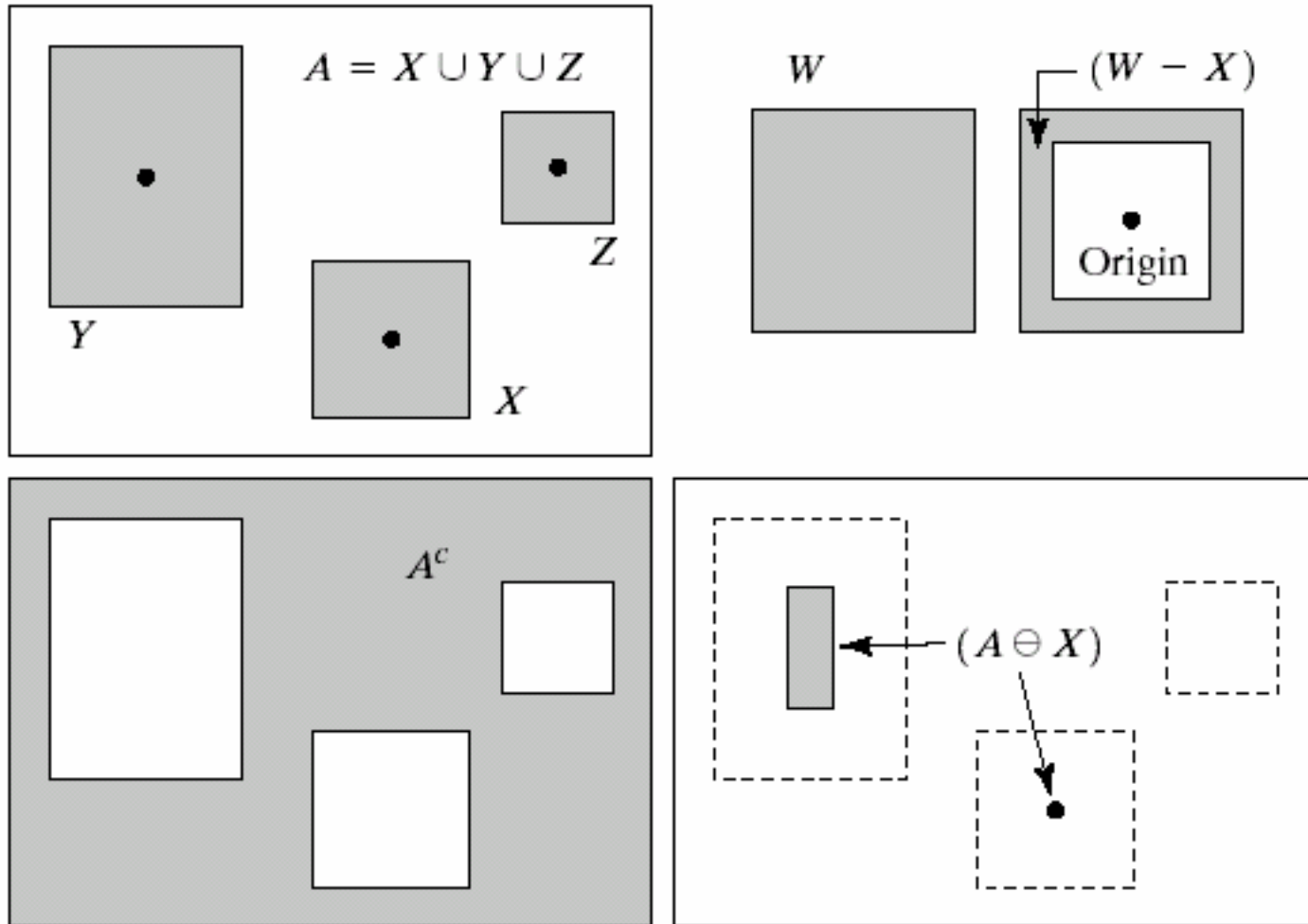
a	b
d	c
e	f

FIGURE 9.11

(a) Noisy image.
 (c) Eroded image.
 (d) Opening of A .
 (e) Dilation of the opening.
 (f) Closing of the opening. (Original image for this example courtesy of the National Institute of Standards and Technology.)

The Hit-or-Miss Transformation

$$A \odot B = (A \ominus X) \cap [A^c \ominus (W - X)] \quad B = (X, W - X)$$



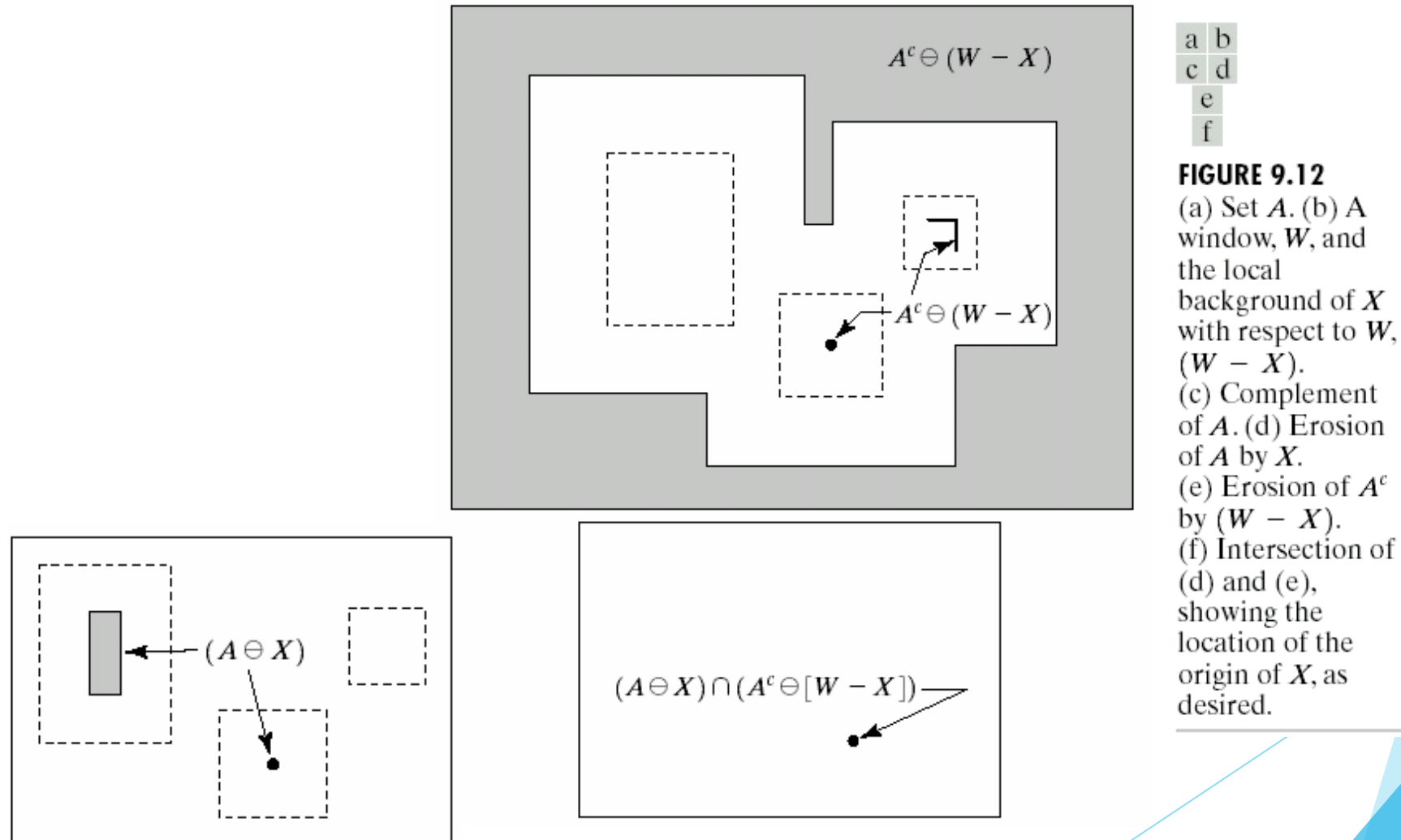
a	b
c	d
e	
f	

FIGURE 9.12

(a) Set A . (b) A window, W , and the local background of X with respect to W , $(W - X)$. (c) Complement of A . (d) Erosion of A by X . (e) Erosion of A^c by $(W - X)$. (f) Intersection of (d) and (e), showing the location of the origin of X , as desired.

The Hit-or-Miss Transformation

$$A \odot B = (A \ominus X) \cap [A^c \ominus (W - X)] \quad B = (X, W - X)$$



Some Basic Morphological Algorithms

Boundary Extraction

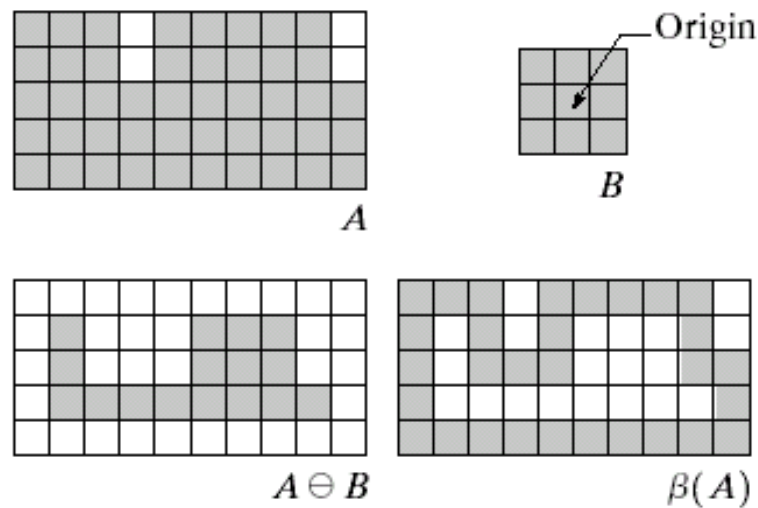
- The boundary of a set A ,

$$\beta(A) = A - (A \ominus B)$$

where B is a suitable structuring element.

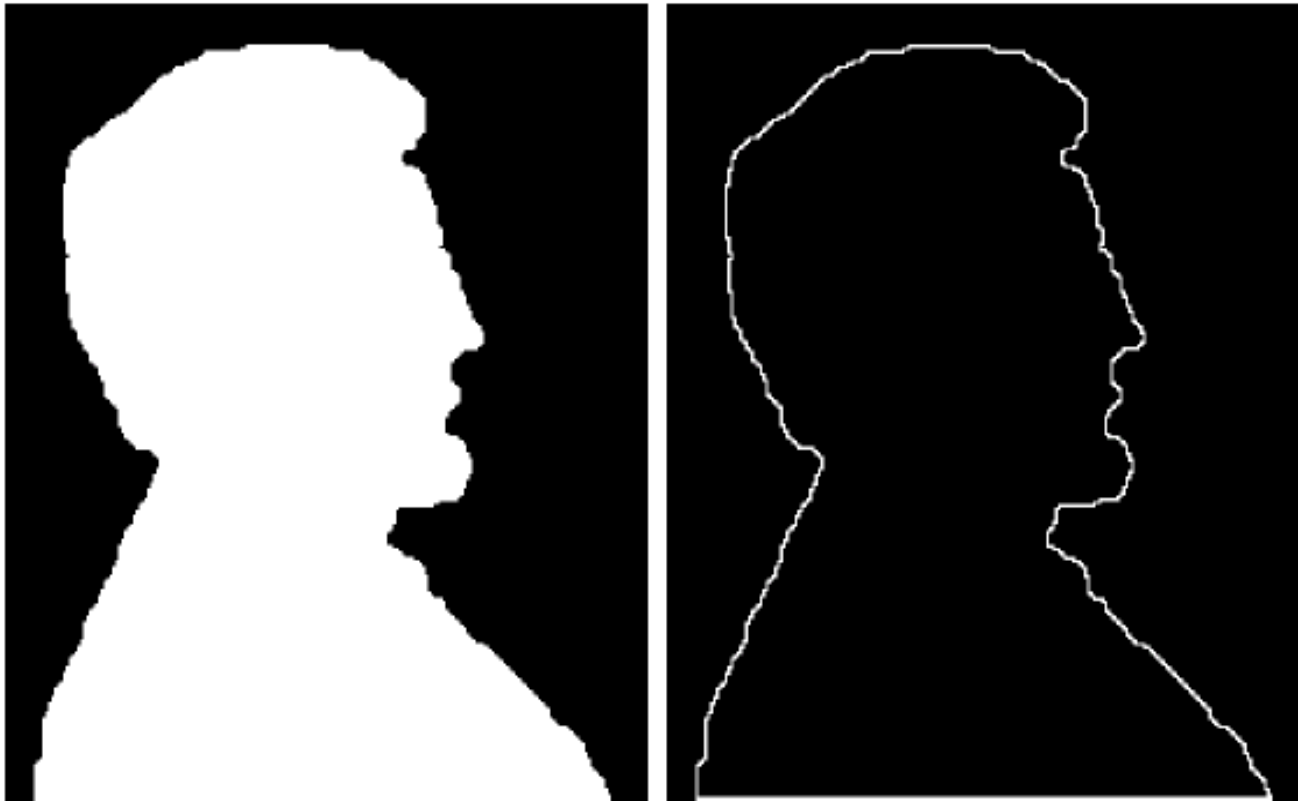
a	b
c	d

FIGURE 9.13 (a) Set A . (b) Structuring element B . (c) A eroded by B . (d) Boundary, given by the set difference between A and its erosion.



Some Basic Morphological Algorithms

Boundary Extraction Example



a b

FIGURE 9.14

(a) A simple binary image, with 1's represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

Some Basic Morphological Algorithms

Region Filling

- ▶ The following procedure can fill the region:

$$X_k = (X_{k-1} \oplus B) \cap A^c$$

where $X_0 = p$, and B is the symmetric structuring element shown in Fig. 9.15.

- ▶ Note that p is the initial point we should assign.

Some Basic Morphological Algorithms

Region Filling

a	b	c
d	e	f
g	h	i

FIGURE 9.15

Region filling.

(a) Set A .

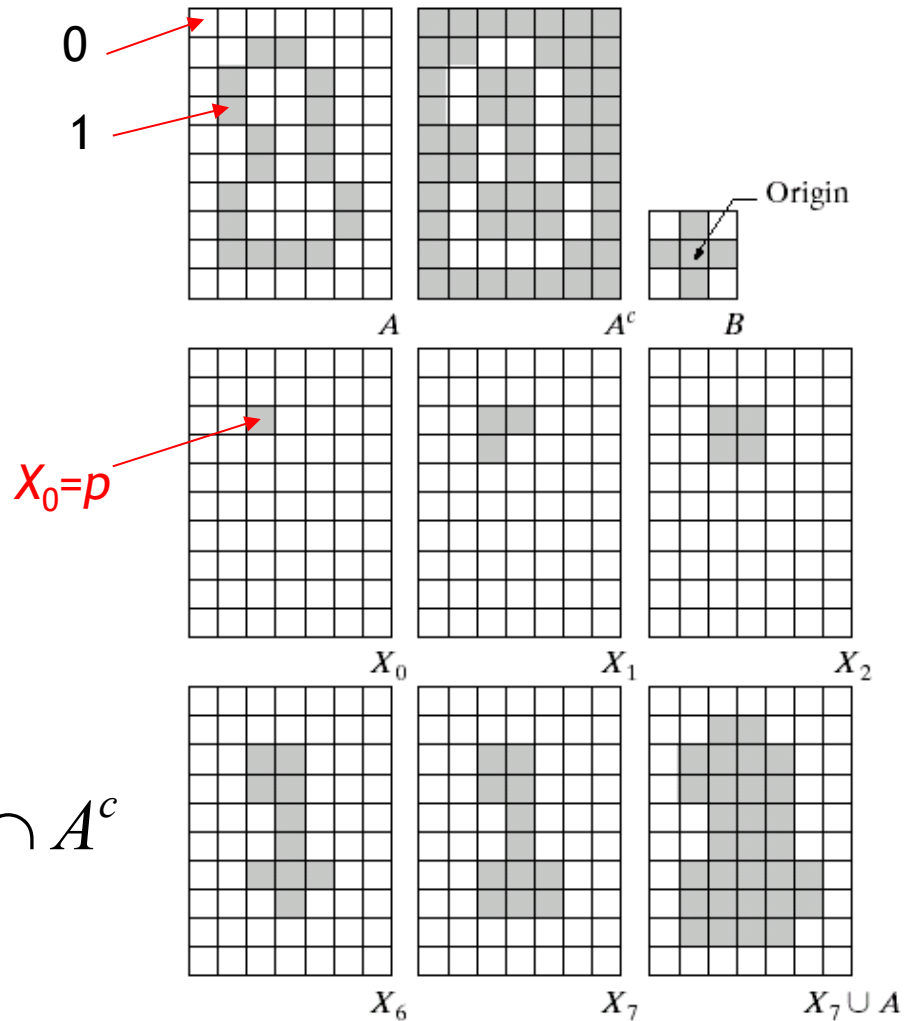
(b) Complement of A .

(c) Structuring element B .

(d) Initial point inside the boundary.

(e)–(h) Various steps of Eq. (9.5-2).

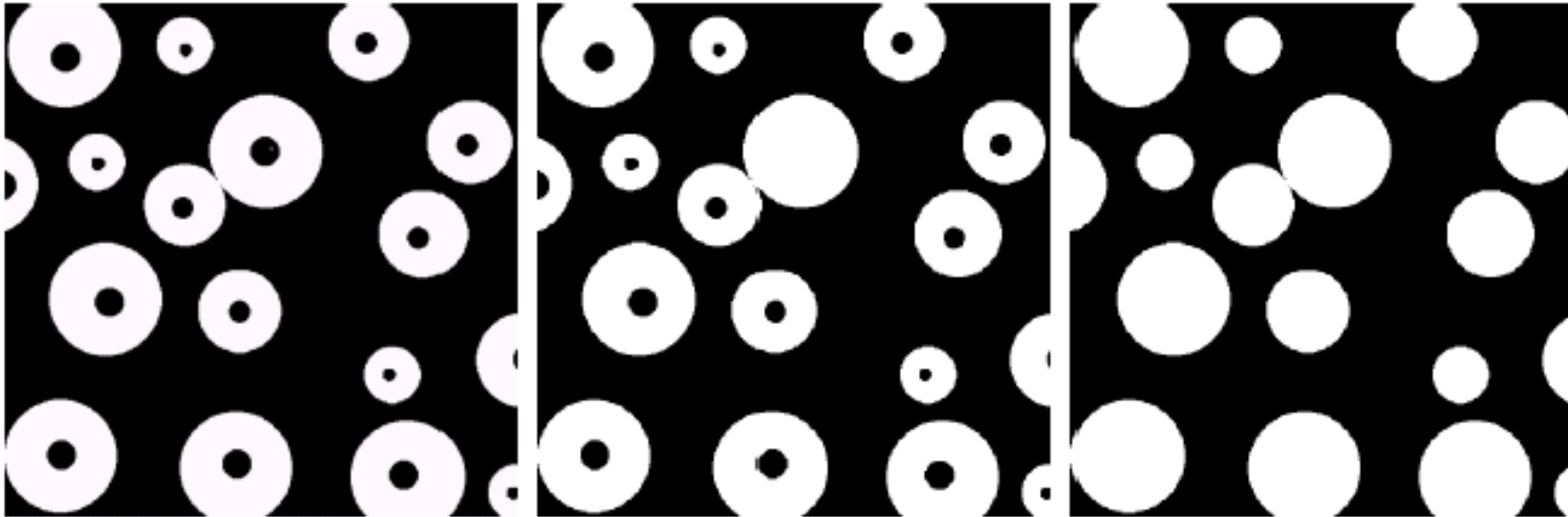
(i) Final result [union of (a) and (h)].



$$X_k = (X_{k-1} \oplus B) \cap A^c$$

Some Basic Morphological Algorithms

Region Filling



a b c

FIGURE 9.16 (a) Binary image (the white dot inside one of the regions is the starting point for the region-filling algorithm). (b) Result of filling that region (c) Result of filling all regions.

Some Basic Morphological Algorithms

Extraction of Connected Components

$$X_k = (X_{k-1} \oplus B) \cap A$$

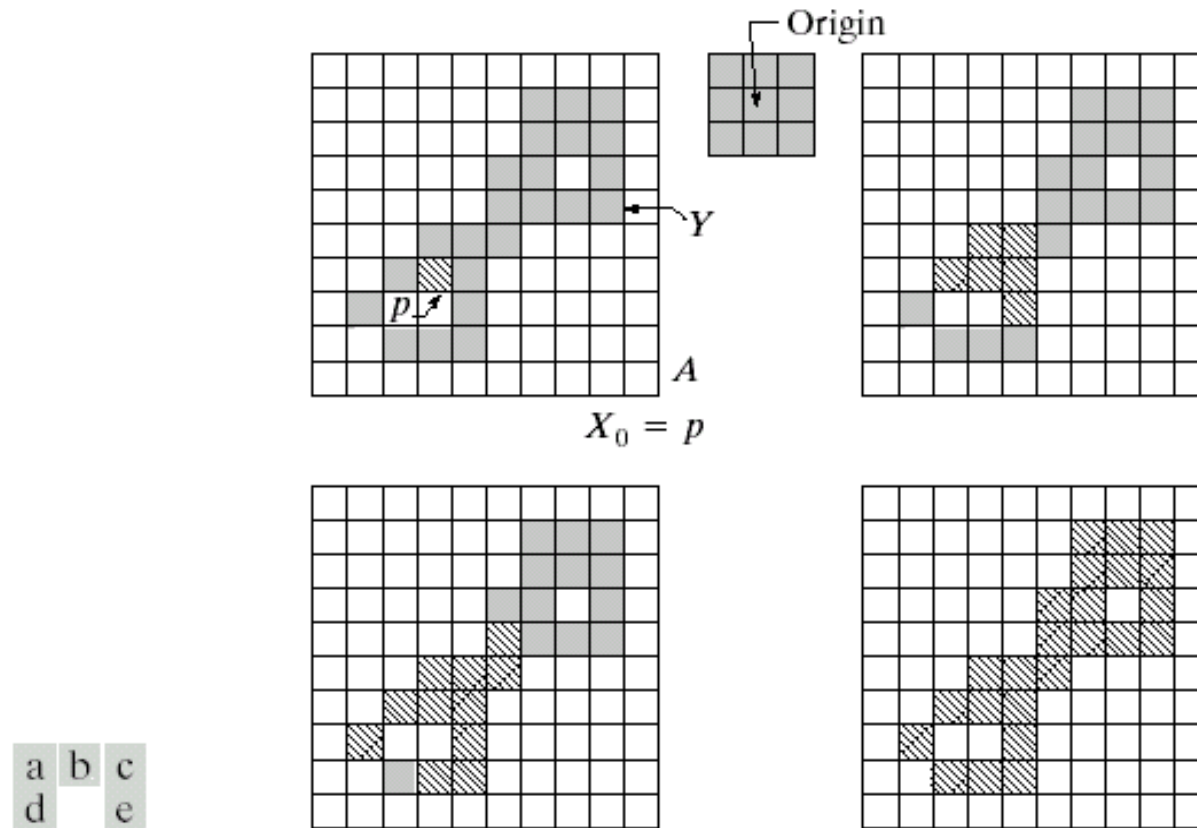


FIGURE 9.17 (a) Set A showing initial point p (all shaded points are valued 1, but are shown different from p to indicate that they have not yet been found by the algorithm). (b) Structuring element. (c) Result of first iterative step. (d) Result of second step. (e) Final result.

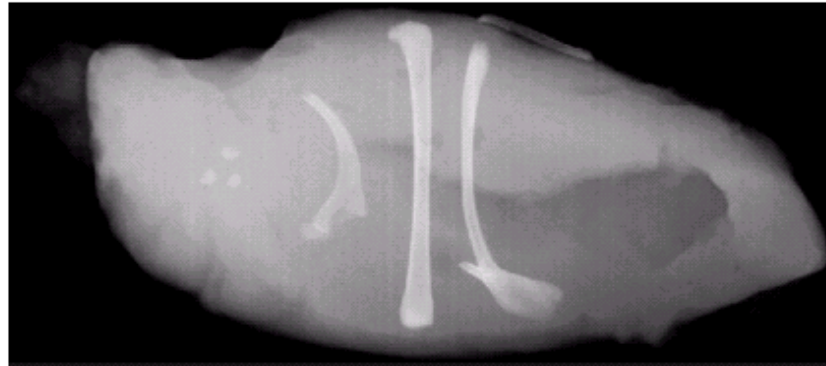
Some Basic Morphological Algorithms

Extraction of Connected Components

a
b
c d

FIGURE 9.18

(a) X-ray image of chicken filet with bone fragments.
(b) Thresholded image. (c) Image eroded with a 5×5 structuring element of 1's.
(d) Number of pixels in the connected components of (c). (Image courtesy of NTB Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)



Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

Some Basic Morphological Algorithms

Thinning

- ▶ The thinning of a set A by a structuring element B can be defined in terms of the hit-or-miss transform:

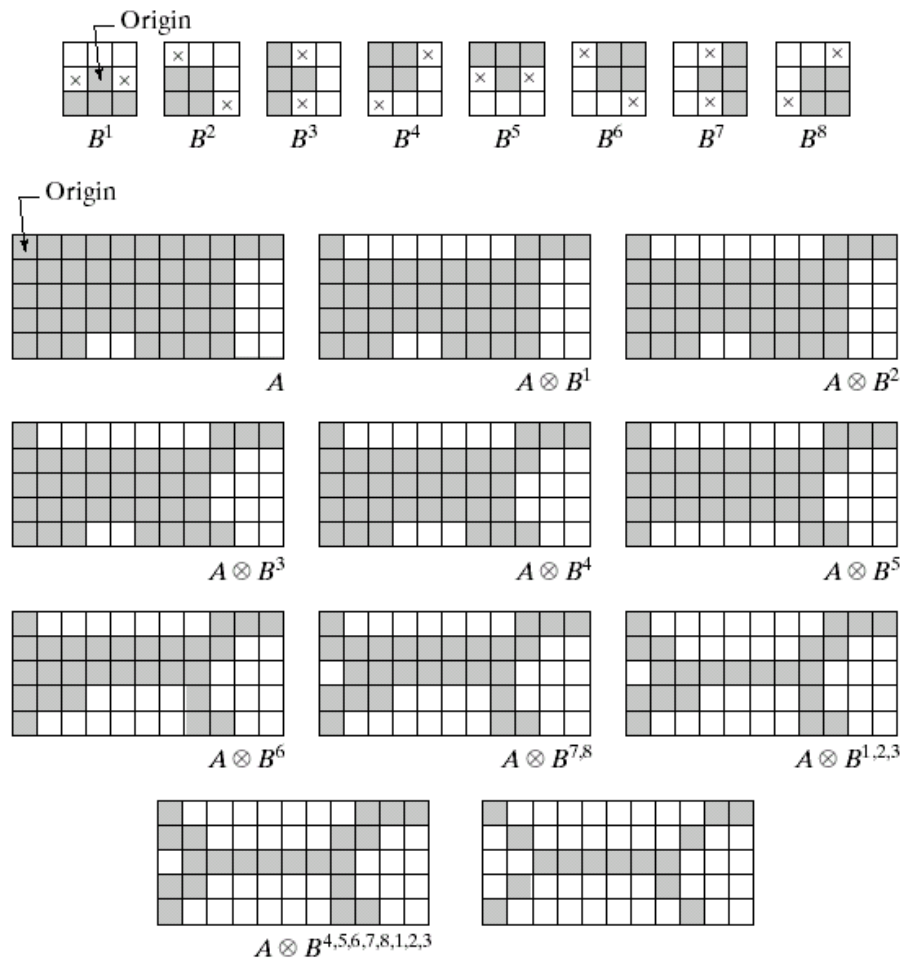
$$\begin{aligned} A \otimes B &= A - (A \odot B) \\ &= A \cap (A \odot B)^c \end{aligned}$$

- ▶ A more useful expression for thinning A symmetrically is based on a sequence of structuring elements:

$$\begin{aligned} \{B\} &= \{B^1, B^2, B^3, \dots, B^n\} \\ A \otimes \{B\} &= ((\dots((A \otimes B^1) \otimes B^2) \dots) \otimes B^n) \end{aligned}$$

Some Basic Morphological Algorithms

Thinning



a
b c d
e f g
h i j
k l

FIGURE 9.21 (a) Sequence of rotated structuring elements used for thinning. (b) Set A . (c) Result of thinning with the first element. (d)–(i) Results of thinning with the next seven elements (there was no change between the seventh and eighth elements). (j) Result of using the first element again (there were no changes for the next two elements). (k) Result after convergence. (l) Conversion to m -connectivity.

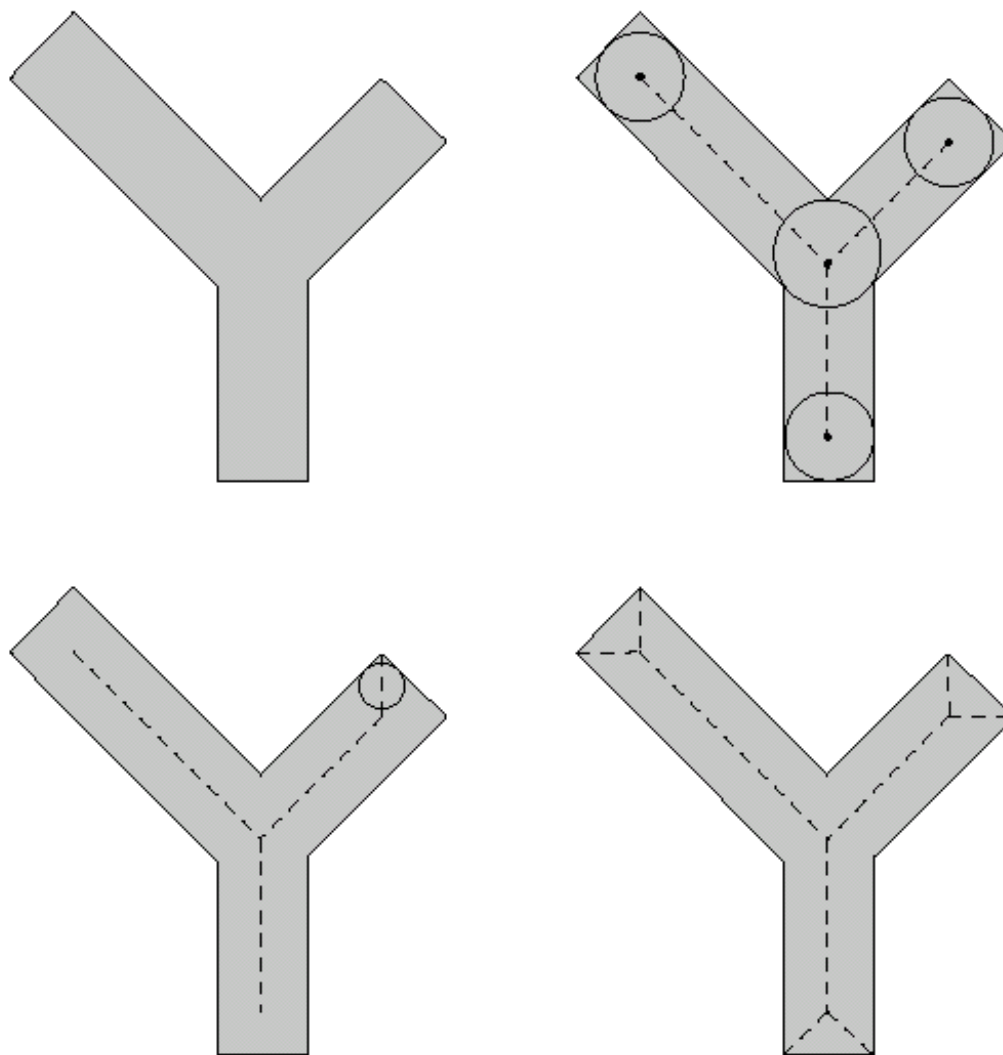
Some Basic Morphological Algorithms

Skeletons

a	b
c	d

FIGURE 9.23

- (a) Set A .
- (b) Various positions of maximum disks with centers on the skeleton of A .
- (c) Another maximum disk on a different segment of the skeleton of A .
- (d) Complete skeleton.



Summary of Morphological Operations and Their Properties

TABLE 9.2

Summary of morphological operations and their properties.

Operation		Equation	Comments (The Roman numerals refer to the structuring elements shown in Fig. 9.26).
Translation		$(A)_z = \{w w = a + z, \text{ for } a \in A\}$	Translates the origin of A to point z .
Reflection		$\hat{B} = \{w w = -b, \text{ for } b \in B\}$	Reflects all elements of B about the origin of this set.
Complement		$A^c = \{w w \notin A\}$	Set of points not in A .
Difference		$A - B = \{w w \in A, w \notin B\}$ $= A \cap B^c$	Set of points that belong to A but not to B .
Dilation		$A \oplus B = \{z (\hat{B})_z \cap A \neq \emptyset\}$	“Expands” the boundary of A . (I)
Erosion		$A \ominus B = \{z (B)_z \subseteq A\}$	“Contracts” the boundary of A . (I)
Opening		$A \circ B = (A \ominus B) \oplus B$	Smooths contours, breaks narrow isthmuses, and eliminates small islands and sharp peaks. (I)
Closing		$A \bullet B = (A \oplus B) \ominus B$	Smooths contours, fuses narrow breaks and long thin gulfs, and eliminates small holes. (I)

Summary of Morphological Operations and Their Properties

Hit-or-miss transform	$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$ $= (A \ominus B_1) - (A \oplus \hat{B}_2)$	The set of points (coordinates) at which, simultaneously, B_1 found a match (“hit”) in A and B_2 found a match in A^c .
Boundary extraction	$\beta(A) = A - (A \ominus B)$	Set of points on the boundary of set A . (I)
Region filling	$X_k = (X_{k-1} \oplus B) \cap A^c; X_0 = p \text{ and } k = 1, 2, 3, \dots$	Fills a region in A , given a point p in the region. (II)
Connected components	$X_k = (X_{k-1} \oplus B) \cap A; X_0 = p \text{ and } k = 1, 2, 3, \dots$	Finds a connected component Y in A , given a point p in Y . (I)
Convex hull	$X_k^i = (X_{k-1}^i \circledast B^i) \cup A; i = 1, 2, 3, 4;$ $k = 1, 2, 3, \dots; X_0^i = A; \text{ and}$ $D^i = X_{\text{conv}}^i.$	Finds the convex hull $C(A)$ of set A , where “conv” indicates convergence in the sense that $X_k^i = X_{k-1}^i$. (III)

Summary of Morphological Operations and Their Properties

Operation	Equation	Comments (The Roman numerals refer to the structuring elements shown in Fig. 9.26).
Thinning	$A \otimes B = A - (A \circledast B)$ $= A \cap (A \circledast B)^c$ $A \otimes \{B\} =$ $((\dots ((A \otimes B^1) \otimes B^2) \dots) \otimes B^n)$ $\{B\} = \{B^1, B^2, B^3, \dots, B^n\}$	Thins set A. The first two equations give the basic definition of thinning. The last two equations denote thinning by a sequence of structuring elements. This method is normally used in practice. (IV)
Thickening	$A \odot B = A \cup (A \circledast B)$ $A \odot \{B\} =$ $((\dots (A \odot B^1) \odot B^2 \dots) \odot B^n)$	Thickens set A. (See preceding comments on sequences of structuring elements.) Uses IV with 0's and 1's reversed.

TABLE 9.2

Summary of morphological results and their properties.
(*continued*)

Summary of Morphological Operations and Their Properties

Skeletons

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

$$S_k(A) = \bigcup_{k=0}^K \{ (A \ominus kB) - [(A \ominus kB) \circ B] \}$$

Reconstruction of A :

$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

Finds the skeleton $S(A)$ of set A . The last equation indicates that A can be reconstructed from its skeleton subsets $S_k(A)$.

In all three equations, K is the value of the iterative step after which the set A erodes to the empty set. The notation $(A \ominus kB)$ denotes the k th iteration of successive erosion of A by B . (I)

Pruning

$$X_1 = A \otimes \{B\}$$

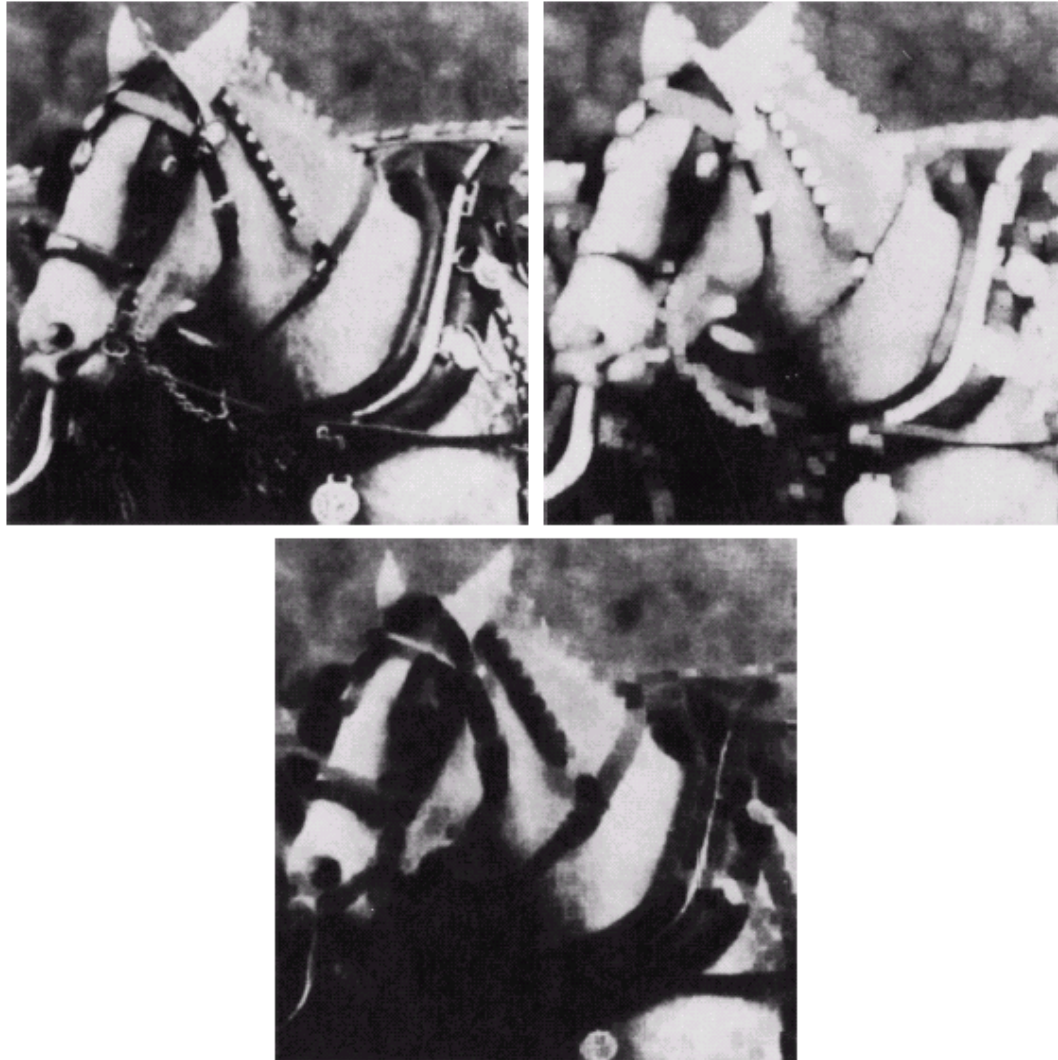
$$X_2 = \bigcup_{k=1}^8 (X_1 \circledast B^k)$$

$$X_3 = (X_2 \oplus H) \cap A$$

$$X_4 = X_1 \cup X_3$$

X_4 is the result of pruning set A . The number of times that the first equation is applied to obtain X_1 must be specified. Structuring elements V are used for the first two equations. In the third equation H denotes structuring element I .

Extensions to Gray-Scale Images



a b
c

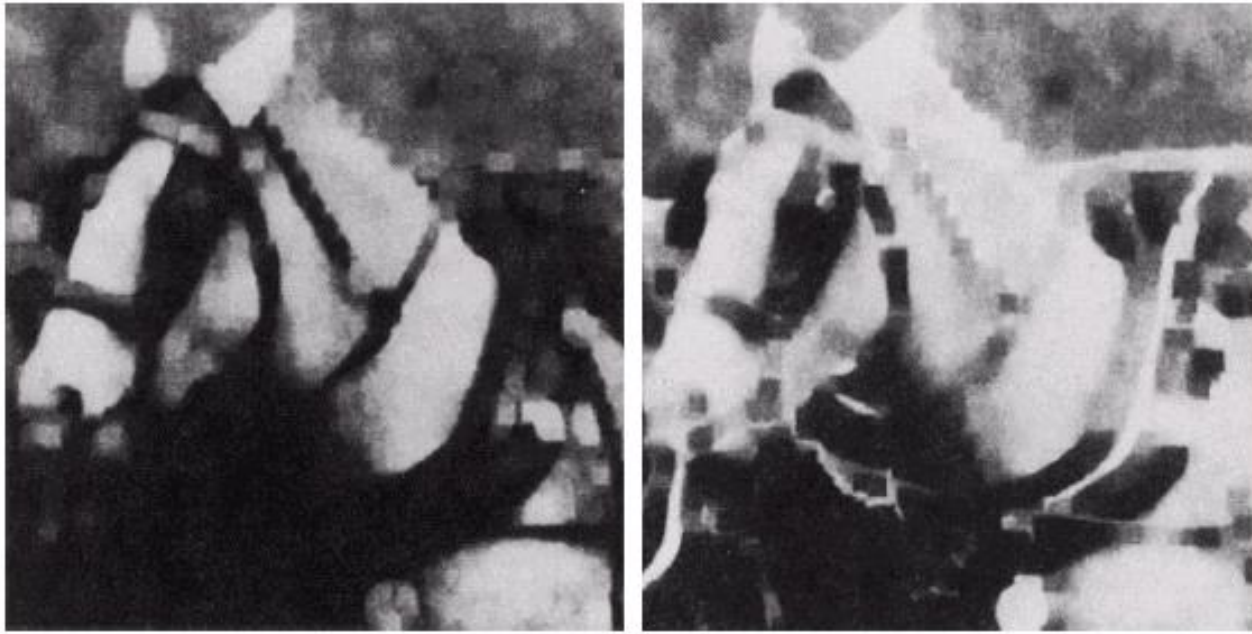
FIGURE 9.29

(a) Original image. (b) Result of dilation.

(c) Result of erosion.

(Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

Extensions to Gray-Scale Images



a b

FIGURE 9.31 (a) Opening and (b) closing of Fig. 9.29(a). (Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

Extensions to Gray-Scale Images

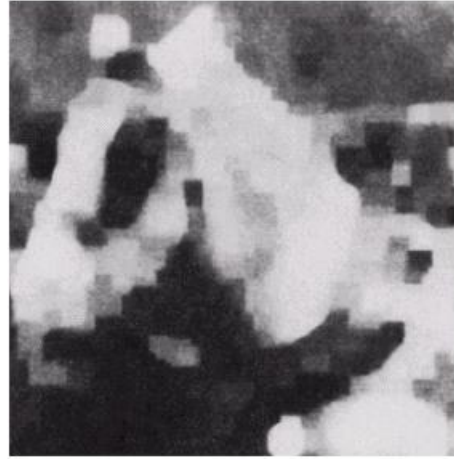


FIGURE 9.32 Morphological smoothing of the image in Fig. 9.29(a). (Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

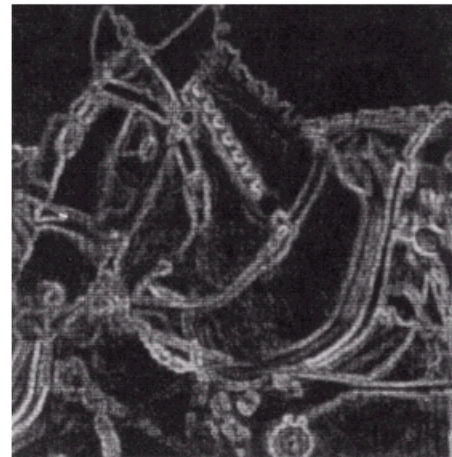


FIGURE 9.33 Morphological gradient of the image in Fig. 9.29(a). (Courtesy of Mr. A. Morris, Leica Cambridge, Ltd.)

Morphological Gradient

The morphological gradient is the difference between the dilation and erosion of an image.

- To highlight the boundaries or edges of objects in the image.
- By subtracting the eroded image from the dilated image, the boundaries of the objects are emphasized.

Top-Hat Transform

The top-hat transform is used to extract small elements and details from an image.

- To enhance bright objects on a dark background or vice versa.
- There are two types: white top-hat (original image minus the result of opening) and black top-hat (result of closing minus the original image)

Skeletonization

Skeletonization reduces objects in a binary image to their skeletal form.

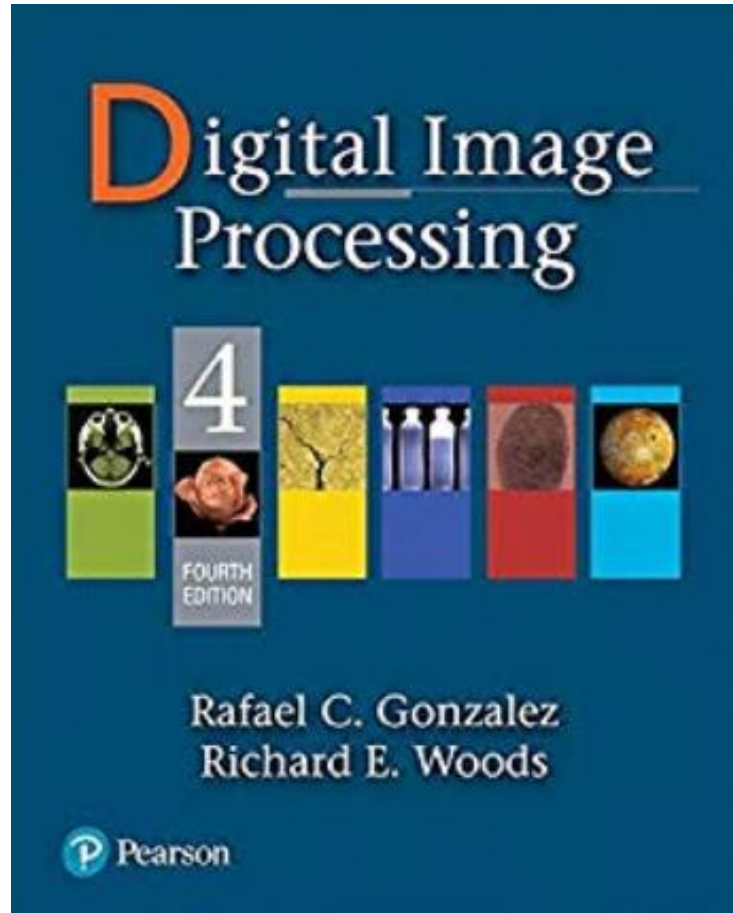
- To simplify objects to their essential structure while preserving their connectivity.
- It iteratively erodes the image until the objects are reduced to a minimal, skeletal form.

Pruning

Pruning removes small spurs or extraneous branches from the skeleton of an image.

- To clean up the skeleton by removing unwanted artifacts.
- It identifies and removes small, irrelevant branches from the skeletonized image.

Reference



Question



Thank you

